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Chapter 4

Series not on TV

4.1 Problem 4.1

Determine whether each of the following converges conditionally, converges absolutely, or diverges. You do not need to prove your answers, but state which of the following tests gives the answers: the k^{th}-term test, the geometric series test, and the alternating series test.

(a) $\sum_{k=1}^{\infty}(-1)^k \frac{1}{\sqrt{k}}$

(b) $\sum_{k=1}^{\infty}(-1)^k \frac{k}{k+7}$

(c) $\sum_{k=1}^{\infty} \frac{1}{(\ln(4))^k}$

(d) $\sum_{k=1}^{\infty} \frac{1}{k!}$

(e) $\sum_{k=1}^{\infty}(\sqrt{k+1}-\sqrt{k})$

(f) $\sum_{k=1}^{\infty} \frac{1}{2}$

(g) $\sum_{k=1}^{\infty}(-1)^k \frac{1}{\sqrt[3]{k}}$

(h) $\sum_{k=53}^{\infty} \frac{\ln(k)}{\ln(\ln(k))}$

(i) $\sum_{k=7}^{\infty} \left(\frac{2}{k} + \frac{3}{k^2}\right)$

4.2 Problem 4.3

Prove that if $\sum_{k=1}^{\infty} |a_k|$ converges, then $\sum_{k=1}^{\infty} a_k$ converge too. This is Propostion 4.18.

4.3 Problem 4.5

Give an example of a series where $\sum_{k=1}^{\infty} a_k$ converges but $\sum_{k=1}^{\infty} a_{2k}$ diverges.

4.4 Problem 4.7

Give an example of a series $\sum_{k=1}^{\infty} a_k$ where:

- $\sum_{k=1}^{\infty} a_k$ converges,
- $\sum_{k=1}^{\infty} a_k^2$ diverges
- $\sum_{k=1}^{\infty} a_k^3$ converges

4.5 Problem 4.9

The geometric series test (Proposition 4.9) says that a geometric series diverges if $|r| \geq 1$. Recall that a series can diverge to ∞, to $-\infty$, or can be "does not exist." Which form of divergence is it for a geometric series? Note that your answer may depend on r.

4.6 Problem 4.11

(a) Find a way to write 77.77777777... as a geometric series, and then prove this number is rational by using the geometric series test to write this number as a fraction with integers in the numerator and denominator.

(b) Write 77.77777777... as a different geometric series, and use the geometric series test to write this number as a fraction with integers in the numerator and denominator. Are your two fractions the same?

(c) A number q has a repeating decimal if the non-integer portion of its decimal expansion is repetitive. For example, 72.578578578578578... has a repeating decimal. Prove that if a number q has a repeating decimal, then q is rational.

4.7 Problem 4.17

Prove the ratio test via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ with $a_k = 0$, assume that

$$\lim_{k \to \infty} \left| \frac{a_{k+1}}{a_k} \right| =: r < 1$$

We will prove that the series converges absolutely.

(a) Let q be such that $r < q < 1$. Explan why there is some N such that $n \geq N$ implies that $|a|_{k+1} \leq |a_k| \cdot q$.

(b) Explain why $\sum_{k=1}^{\infty} |a_N| \cdot q^k$ necessarily converges.

(c) Finally, use part (b) to prove that $\sum_{k=1}^{\infty} |a_k|$ converges.

4.8 Problem 4.18

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